NATIONAL UNIVERSITY OF PAKISTAN
Department of Computer Science

TRAFFIC LIGHT CONTROLLER

Digital Logic Design (DLD) – Semester Project Report

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Course: Digital Logic Design

Simulation Platforms: Tinkercad

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1. Introduction

A Traffic Light Controller is a classic sequential logic system used to automatically regulate the flow of vehicles at road intersections by cycling through Red, Yellow, and Green signals in a fixed, repeating sequence. This project implements a basic single-direction Traffic Light Controller as a semester project for the Digital Logic Design (DLD) course, using an Arduino Uno to control three LEDs that represent the Stop, Caution, and Go signals.

The project was implemented and simulated on Tinkercad, using a 3D circuit model and a microcontroller-based circuit simulation, demonstrating both the physical hardware layout and the timing logic that drives the LED sequence.

2. Objectives

1. To design and simulate a working Traffic Light Controller circuit.
2. To understand and apply sequential timing logic using an Arduino microcontroller.
3. To control multiple LEDs through digital output pins in a fixed time sequence.
4. To gain practical exposure to circuit simulation tools (Tinkercad) and 3D circuit modelling.
5. To translate the real-world traffic signal sequence (Red → Green → Yellow → Red) into a digital logic / microcontroller program.

3. Components Used

Component	Quantity	Purpose
Arduino Uno	1	Main microcontroller board that runs the control logic
LED – Red	1	Indicates STOP signal
LED – Yellow/Orange	1	Indicates CAUTION signal
LED – Green	1	Indicates GO signal
Resistors (220Ω)	3	Current-limiting resistors for each LED
Breadboard	1	Platform for assembling the circuit
Jumper Wires	Multiple	Electrical connections between Arduino and breadboard
USB Cable	1	Power supply and code upload from computer

4. 3D Circuit Model

The figure below shows the 3D model of the Traffic Light Controller as designed in Tinkercad. It represents the physical appearance of a real traffic signal housing with three coloured lights — Red, Yellow, and Green — mounted on a pole, giving a realistic visualization of the final outcome of the project.

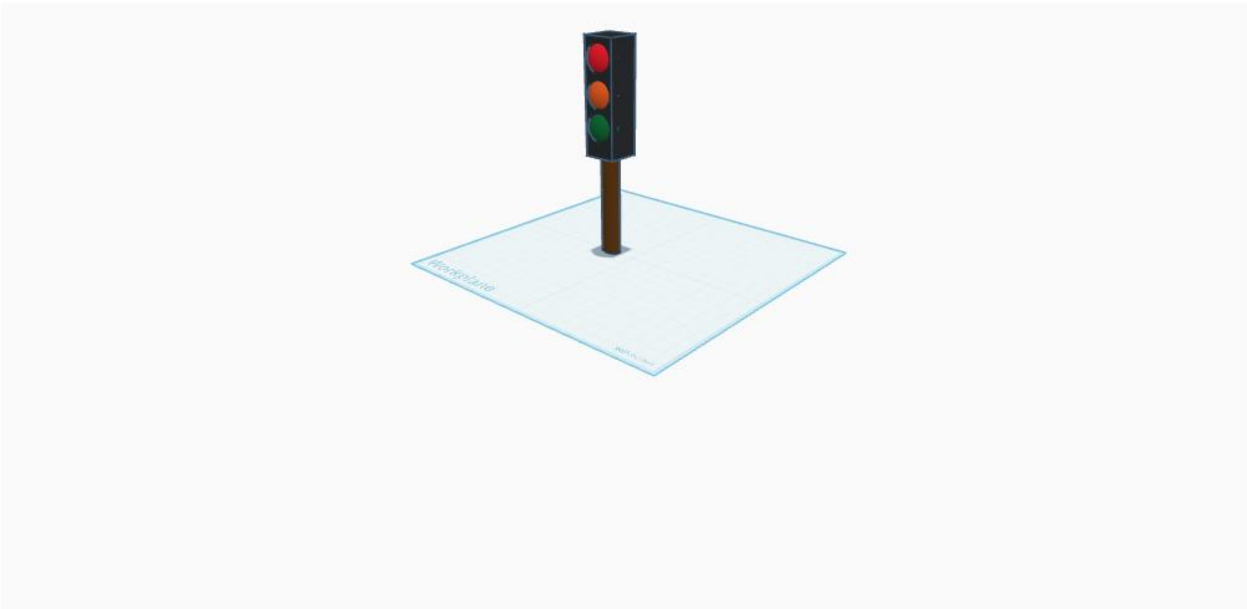

Figure 1: 3D model of the Traffic Light (Tinkercad)

5. Circuit Diagram

The circuit was built using an Arduino Uno connected to a breadboard. Three LEDs (Green, Yellow, and Red) are each connected in series with a 220Ω current-limiting resistor, and wired to three separate digital output pins of the Arduino, as shown below.

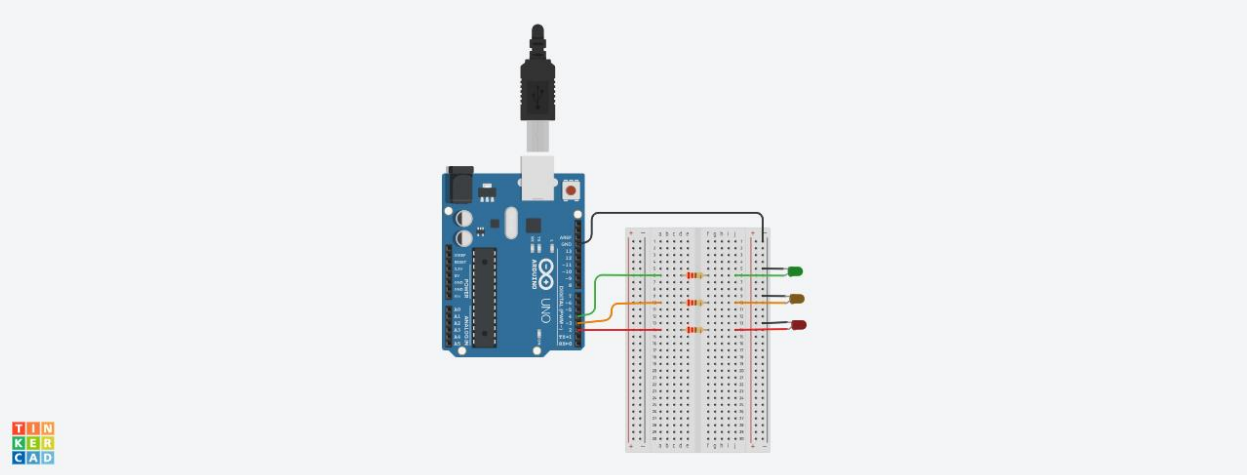

Figure 2: Circuit diagram – Arduino Uno with breadboard and LEDs (Tinkercad)

5.1 Pin Configuration

Arduino Pin	Component	Function
Pin 2	Red LED	Stop signal — ON for 5 seconds
Pin 3	Orange/Yellow LED	Caution signal — ON for 2 seconds
Pin 4	Green LED	Go signal — ON for 5 seconds

6. Working Principle

The Arduino is programmed to switch each LED ON and OFF in a fixed, repeating sequence using the digitalWrite() and delay() functions:

- The Green LED turns ON for 5 seconds, signalling vehicles to GO.
- The Green LED turns OFF, and the Yellow/Orange LED turns ON for 2 seconds, signalling vehicles to slow down (CAUTION).
- The Yellow LED turns OFF, and the Red LED turns ON for 5 seconds, signalling vehicles to STOP.
- The Red LED turns OFF, and the cycle repeats continuously inside the loop() function.

This timing sequence mimics the real behaviour of a traffic signal at a road intersection, with appropriately longer durations for Go and Stop, and a shorter duration for the Caution signal.

7. Arduino Source Code (.ino)

The complete C++ code uploaded to the Arduino Uno to control the LED sequence is given below:

```
void setup()
{
  pinMode(4, OUTPUT); //For Green LED
  pinMode(3, OUTPUT); //For Orange/Yellow LED
  pinMode(2, OUTPUT); //For Red LED
}
void loop(){
  digitalWrite(4, HIGH); //Turn ON Green LED
  delay(5000); //wait for 5 seconds
  digitalWrite(4, LOW); //Turn off Green LED

  digitalWrite(3, HIGH); //Turn ON Orange/Yellow LED
  delay(2000); //wait for 2 seconds
  digitalWrite(3, LOW);  //Turn off Orange/Yellow LED

  digitalWrite(2, HIGH); //Turn ON Red LED
  delay(5000); //wait for 5 seconds
  digitalWrite(2, LOW); //Turn off Red LED
}
```

7.1 Code Explanation

- setup(): Runs once when the Arduino starts. It configures pins 2, 3, and 4 as OUTPUT pins for the Red, Yellow, and Green LEDs respectively.
- loop(): Runs continuously and repeatedly executes the Green → Yellow → Red sequence with the specified delay() timings (in milliseconds).
- digitalWrite(pin, HIGH/LOW): Turns the specified LED ON (HIGH) or OFF (LOW).
- delay(ms): Pauses program execution for the given number of milliseconds, controlling how long each light stays ON.

8. Simulation Results

The circuit was successfully simulated on Tinkercad. On running the simulation, the LEDs lit up in the correct order and for the correct durations as defined in the code: Green for 5 seconds, Yellow for 2 seconds, and Red for 5 seconds, repeating continuously. This confirmed that both the hardware connections and the microcontroller logic were functioning as intended.

9. Conclusion

This project successfully demonstrates the design, simulation, and working of a basic single-direction Traffic Light Controller using an Arduino Uno. It applies core Digital Logic Design concepts such as sequential timing control and digital output switching, and provides hands-on experience with circuit simulation tools. The project fulfils the requirements of the DLD semester project by combining a realistic 3D model, a functional circuit diagram, and working microcontroller code.

10. Future Enhancements

- Extend the design to a 4-way intersection with synchronized signals on all sides.
- Add a pedestrian crossing signal with its own timing logic.
- Implement the same logic using pure digital logic gates and flip-flops (sequential circuit) instead of a microcontroller, for direct DLD hardware demonstration.
- Add a countdown display (7-segment display) showing the remaining seconds for each signal.